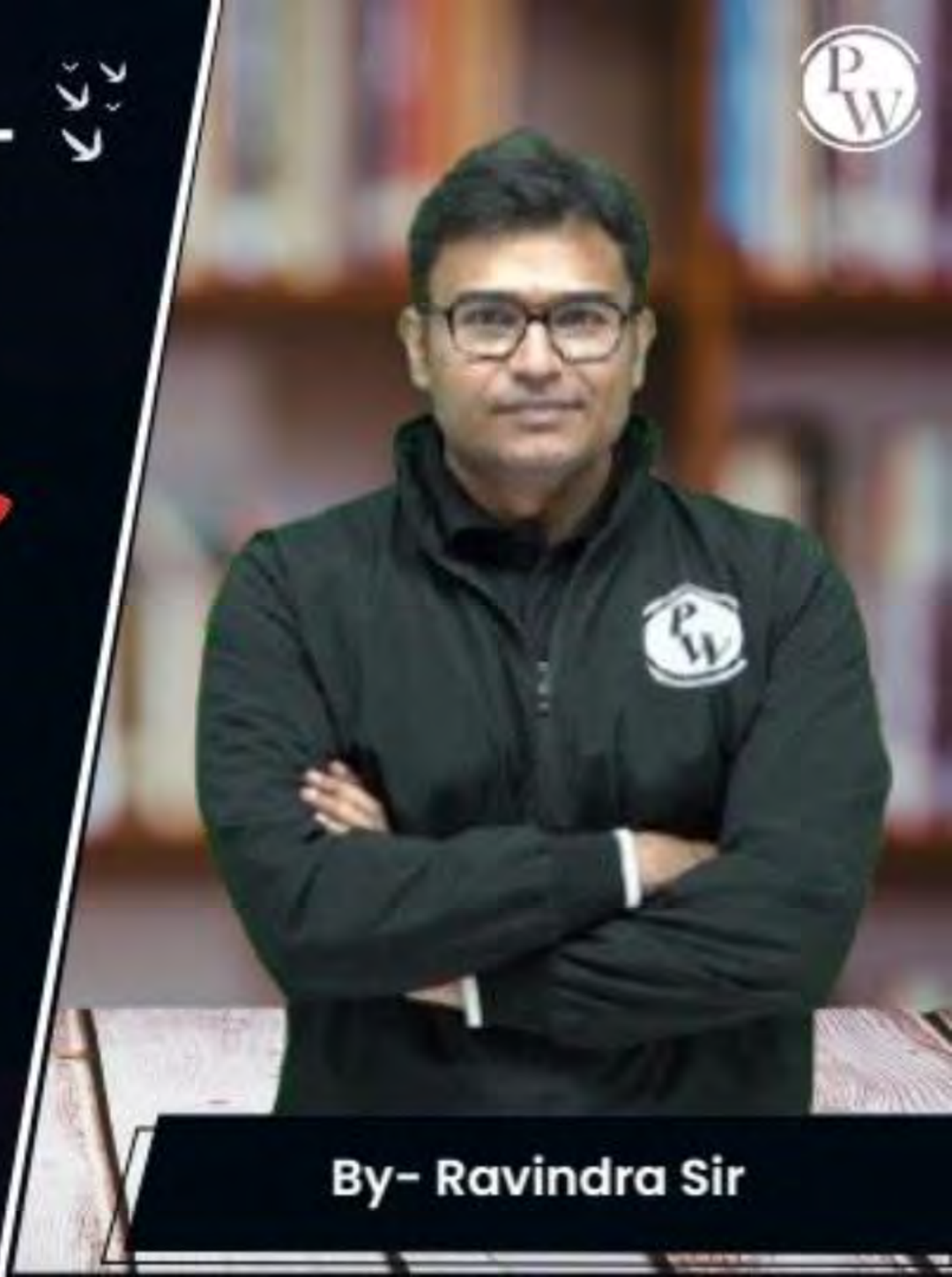
Computer Science & IT

Theory of Computation



Regular Languages

Lecture No. 12



By- Ravindra Sir

Recap of Previous Lecture



Topic

Topic

Topic

Regular grammars

Topics to be Covered



- Topic
- Topic
- Topic

Reg grammar
algorithm in Regular languages

$\Sigma = \{a, b\}$

XFA XRG

Reversal

XReg

$ww^R / w \in (a,b)^*$

✓CFL

$w = abb$

$w^R = bba$

$ww^R = abbbba$

Even length palindrome

Palindrom

$w = aab$

$w^R = baa$

$ww^R = aabbaa$

$\check{a}\,b\check{b}\check{b}\check{b}\check{a}$

CFG

$S \to \check{a}S\check{a} / \check{b}S\check{b} / \epsilon$



$\Sigma = \{a, b\}$



DCFL → Deterministic Context free language

CFL → Context free

CSL → Context sensitive

Rec – Recursive

RE → Recursively enumerable

$\Sigma = \{a, b\}$

$L = \{ wawR \cup wbwR / w \in (a,b)^* \}$

$wawR \rightarrow$ $wbwR$

odd length palindrome

$w = aaab$

aab(a)baa

ϵ

× Reg × RG × FA

VCFL

CFG

$S \rightarrow aSa / bSb / a / b / \epsilon$



abaaba

All palindromes

$S \rightarrow aSa / bSb / a / b / \epsilon$

odd

even

$a^n b^n c^m / n, m \geq 1$ XReg

✓CFL
XReg

$S \rightarrow AB$

$A \rightarrow aAb / ab$ } CFG

$B \rightarrow cB / c$

$a^n c^m b^n / m,n \geq 1$

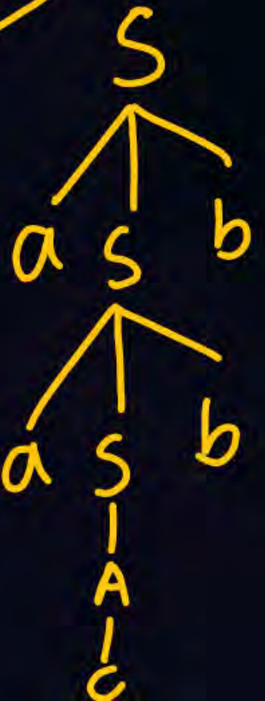
~~abc~~ acb ✓

$S \to aSb / aAb$

$A \to cA / c$

~~$a^* b^* c$~~

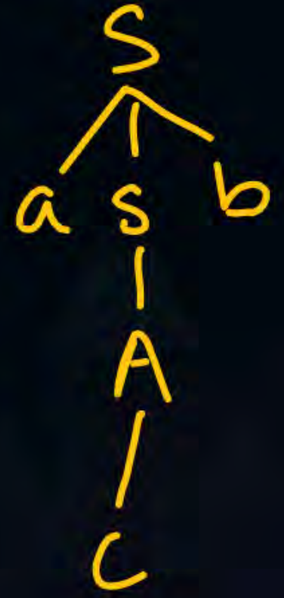
$a^* c^* b^*$





$S \to aSb / aAb$

$A \to cA / c$



$a^n b^n c^m d^m / n, m \geq 1$

$S \rightarrow AB$

$A \rightarrow aAb / ab$

$B \rightarrow cBd / cd$

$a^n b^n c^n / n \geq 1$

Reg X

CFG X → one comparison

CSL ✓

Reg ? ✓

a) b) c) d) ✓

$\overline{a^n b^{2n}} / n \geq 1$ Reg X

CFL ✓

$S \rightarrow aSbb / abb$

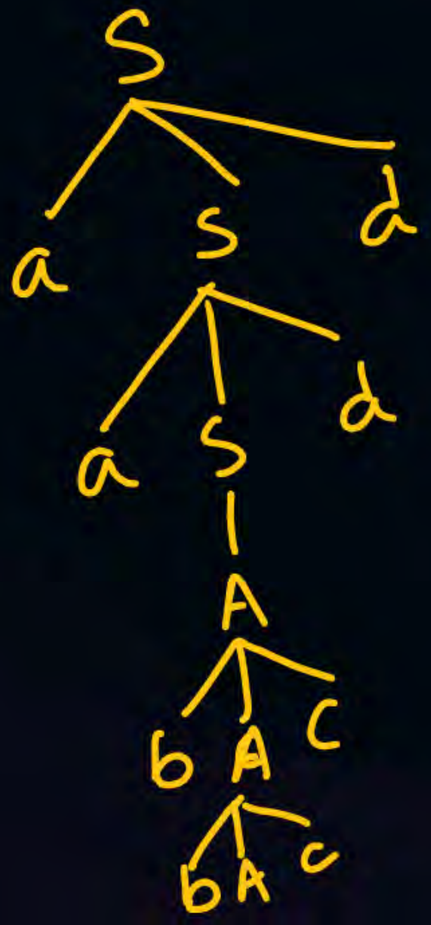


$a^n b^m c^m d^n / n, m \geq 1$ min abcd

$S \rightarrow aSd / aAd$

$A \rightarrow bAc / bc$





$a^n b^m c^n d^m / m, n \geq 1$

✗ Reg ✗

✗ CFG ✗

✓ CSL

$a^{m+n} b^m c^n / n, m \geq 1$ ✗ Reg

$a^n a^m b^m c^n / n, m \geq 1$

$a^m a^n b^m c^n$

$S \to aSc / aAc$

$A \to aAb / ab$



$a^n b^{m+n} c^m / n, m \geq 1$

$a^n b^n b^m c^m$

Reg ✗

CFL ✓

$S \rightarrow AB$

$A \rightarrow aAb/ab$

$B \rightarrow bBc/bc$

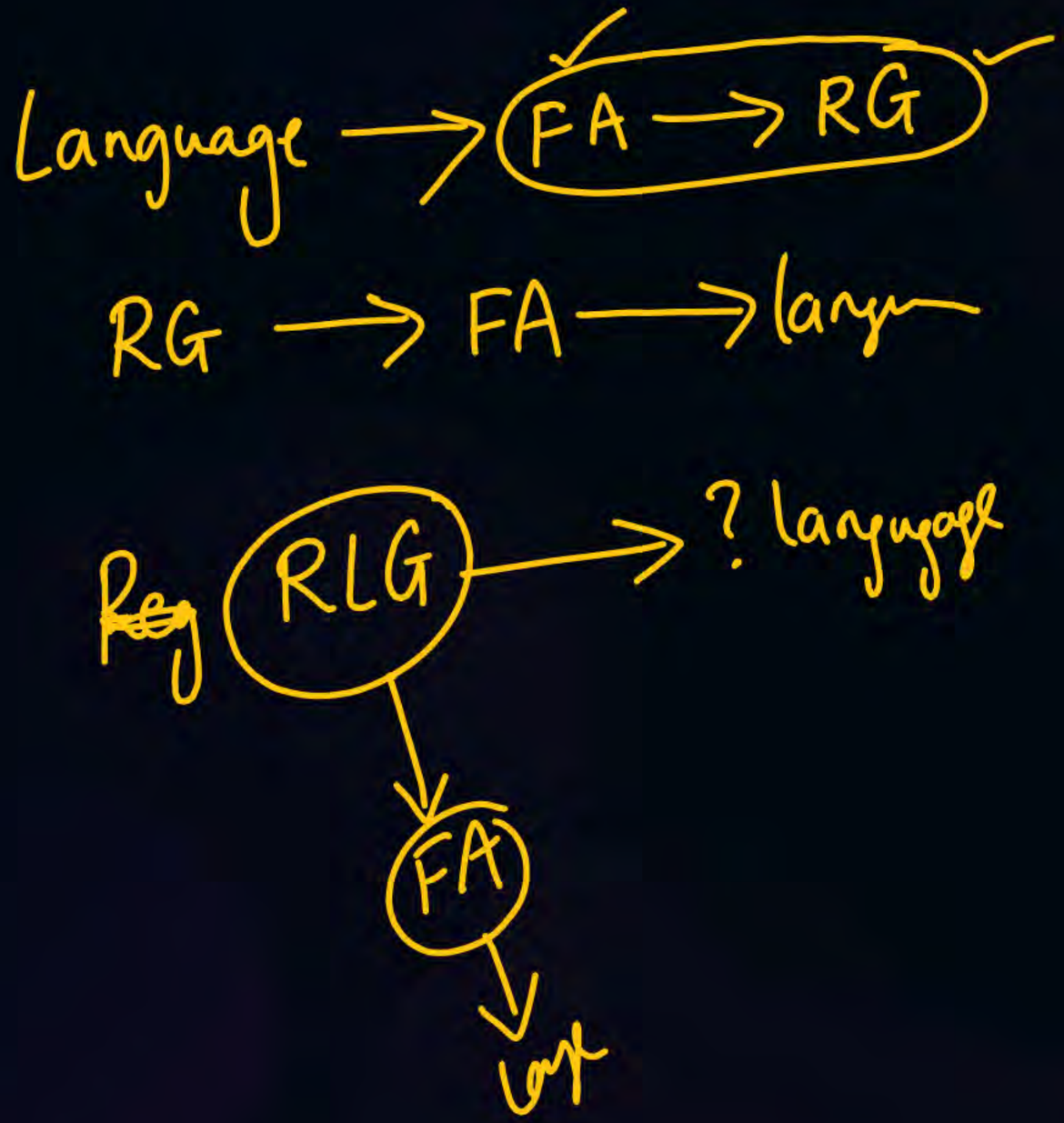
→ CFG

$a^n b^m c^{m+n} / n, m \geq 1$

$a^n b^m c^m c^n$

$S \to aSc / aAc$

$A \to bAc / bc$


Language → FA → RG
RG → FA → lang
Reg RLG → ? language
FA
lang

Rules to Convert RLG to FA :-

$A \to \epsilon$



$A \to a$



$A \to ab$



$A \to aB$



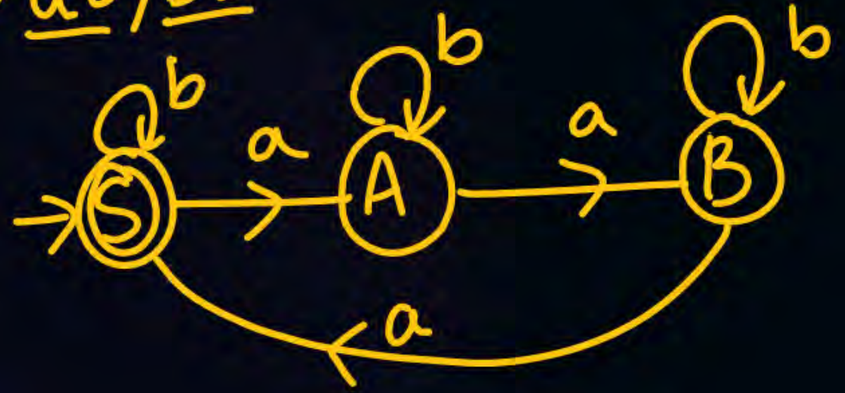
What Lenyuge is generated by following grammar

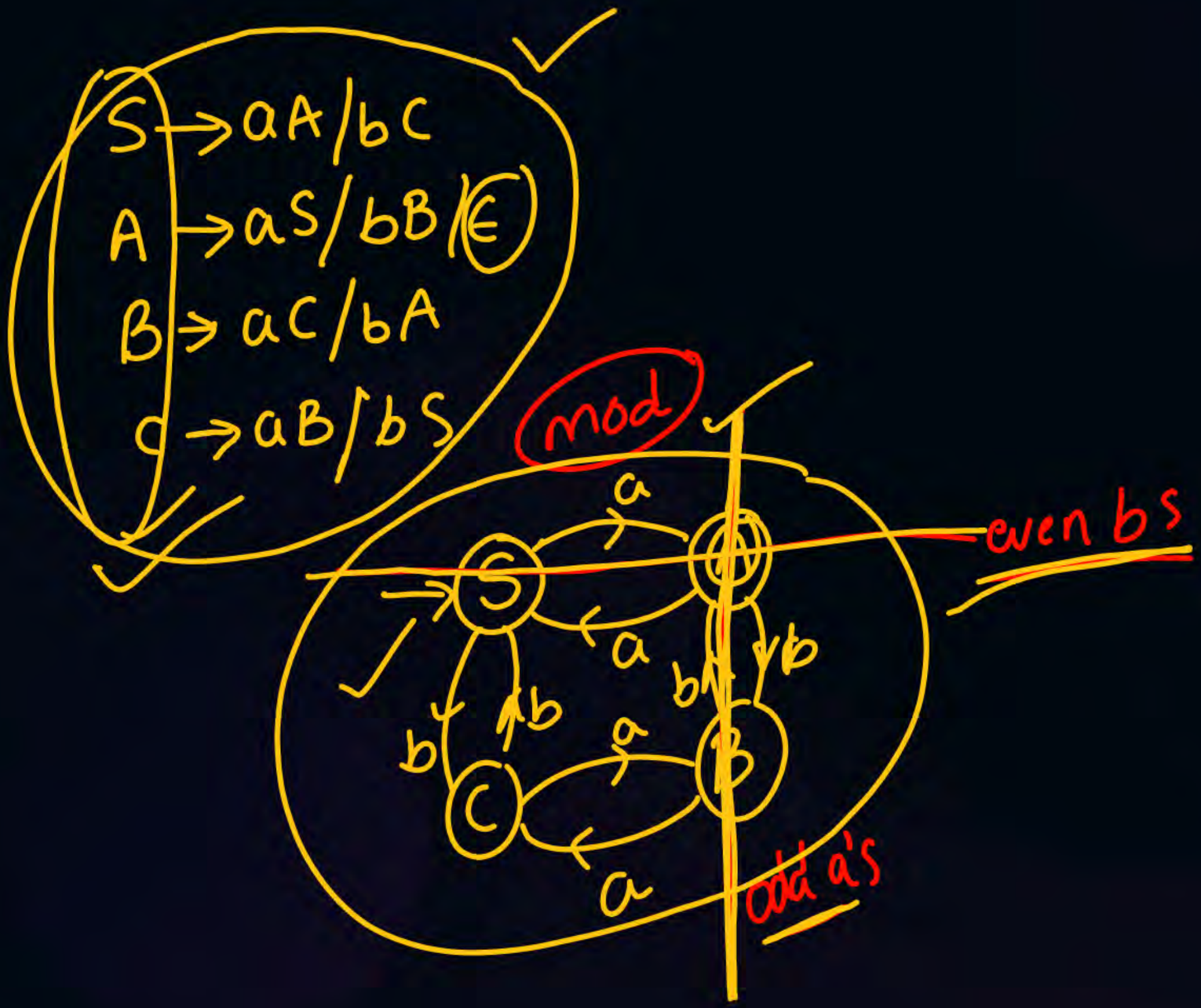
$S \to \underline{aA} / \underline{bS} / \epsilon$

$A \to \underline{aB} / \underline{bA}$

$B \to \underline{aS} / \underline{bB}$

$\Longrightarrow$ FA




S → aA/bC
A → aS/bB/∈
B → aC/bA
C → aB/bS
mod
even bs
odd a's
S
A
B
C
a
b

RLG → FA

LLG —reverse→ RLG → FA —reverse→ FA

(L) (L^R) (L^R) (L^R)^R = L



RLG

$A \to aB$

$B \to aB/bB/\in$



Starting with a

Reverse (Ini ↔ final, trans reverse)



(L^R) → ending with a

$A \to \in$

$B \to aB/bB/aA \to (L^R)$

Rem

$A \to \in$

$B \to Ba/Bb/Aa$ $(L^R)^R = L$

FA (L)

↓

RLG (L)

FA(L)

↓ rev

FA(L^R)

↓

RLL((L^R))

↓ rev

LLG $(L^R)^R = L$.

DFA
NFA
RE
Reg Gram
CFG

Algorithms in reg languages :-

1) NFA → DFA
2) DFA → min DFA
3) ∈ NFA → NFA
4) mealy → moore
5) moore → mealy

$O(n^k)$ polynomial time

X6) NFA → min NFA → non polynomial

intractable

If a problem has an algorithm, then it is called decidable

If the algo is polynomial, then it is called tractable

∈ NFA → min DFA ?

∈ NFA → NFA → DFA → min DFA

NFA → DFA (Subset Construction method)

	a	b
→ q_0	q_0, q_2	—
* q_1	q_2	q_1
q_2	q_0	q_1

	a	b
→ $[q_0]$	$[q_0 q_2]$	D
D	D	D
$[q_0 q_2]$	$[q_0 q_2]$	$[q_1]$
* $[q_1]$	$[q_2]$	$[q_1]$
$[q_2]$	$[q_0]$	$[q_1]$

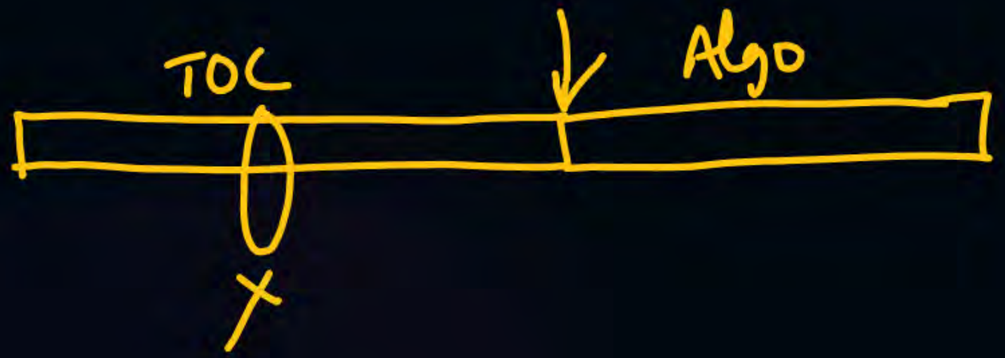
NFA → DFA

DFA

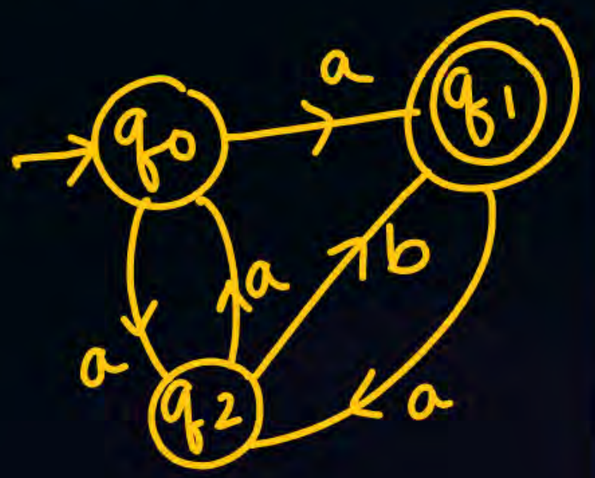
	a	b
$\rightarrow q_0$	q_1, q_2	—
$*q_1$	q_2	q_1
q_2	q_0	q_1

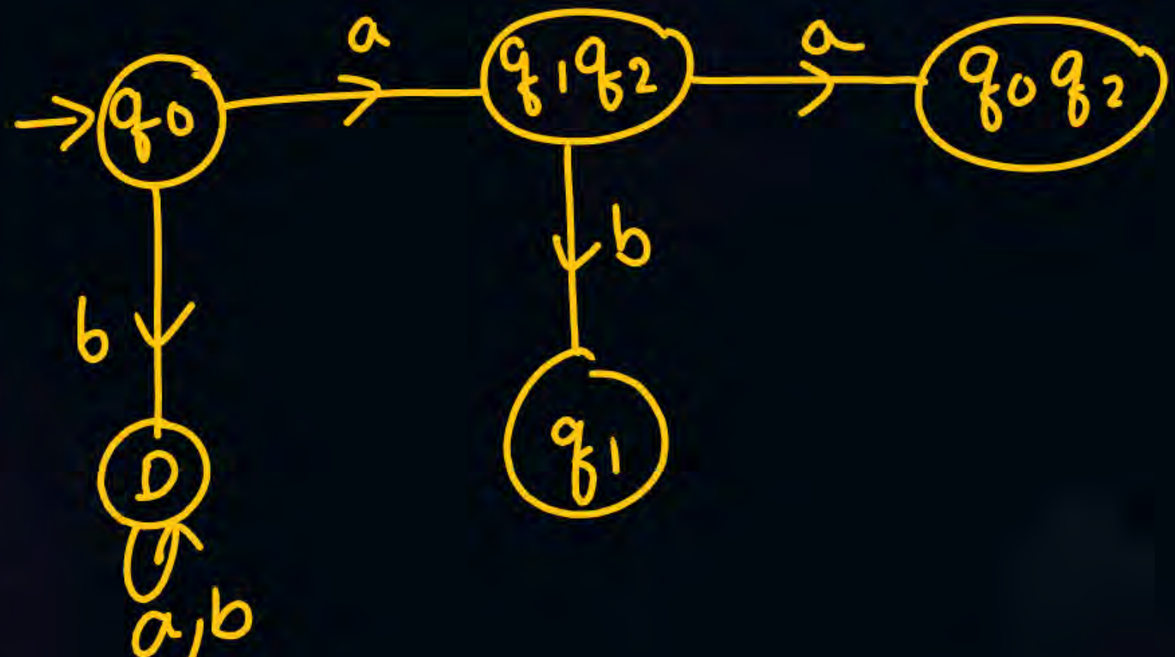
	a	b
$\rightarrow [q_0]$	$[q_1 q_2]$ ✓	D ✓
D	D	D
$*[q_1 q_2]$	$[q_0 q_2]$ ✓	$[q_1]$ ✓
$*[q_1]$	$[q_2]$ ✓	$[q_1]$ ✓
$[q_2]$	$[q_0]$ ✓	$[q_1]$ ✓
$[q_0 q_2]$	$[q_0 q_1 q_2]$ ✓	$[q_1]$ ✓
$*[q_0 q_1 q_2]$	$[q_0 q_1 q_2]$ ✓	$[q_1]$ ✓

When NFA is Converted to DFA, the DFA may not be minimal


TOC
Algo
X

NFA to DFA





$q_0 \xrightarrow{a} q_1q_2$

$q_0 \xrightarrow{b} D$

$q_1 \xrightarrow{a} q_2$

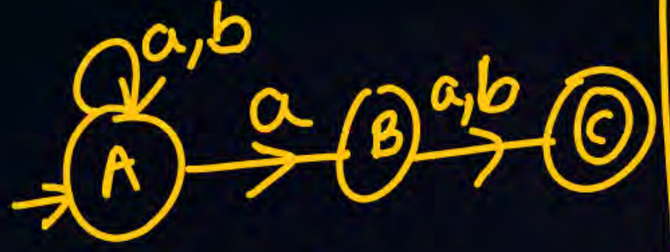
$q_2 \xrightarrow{} q_0$

$q_1 \xrightarrow{b} q_1$

$q_2 \xrightarrow{} q_1$

$\Sigma = \{a, b\}$ NFA L = Second Symbol from RHS is 'a'

$$\frac{(a+b)^* a (a+b)}{(a+b)^* (aa + ab)}$$



✓

2nd Symbol from RHS

	a	b
→A	A,B	A
B	C	C
*C	–	–

NFA – 3
DFA – 4

	a	b
→[A]	[AB] ✓	[A] ✓
[AB]	[ABC] ✓	[AC] ✓
*[AC]	[AB] ✓	[A] ✓
*[ABC]	[ABC] ✓	[AC] ✓

min DFA

$\Sigma = \{a, b\}$ L = 3rd symbol from RHS is 'a'



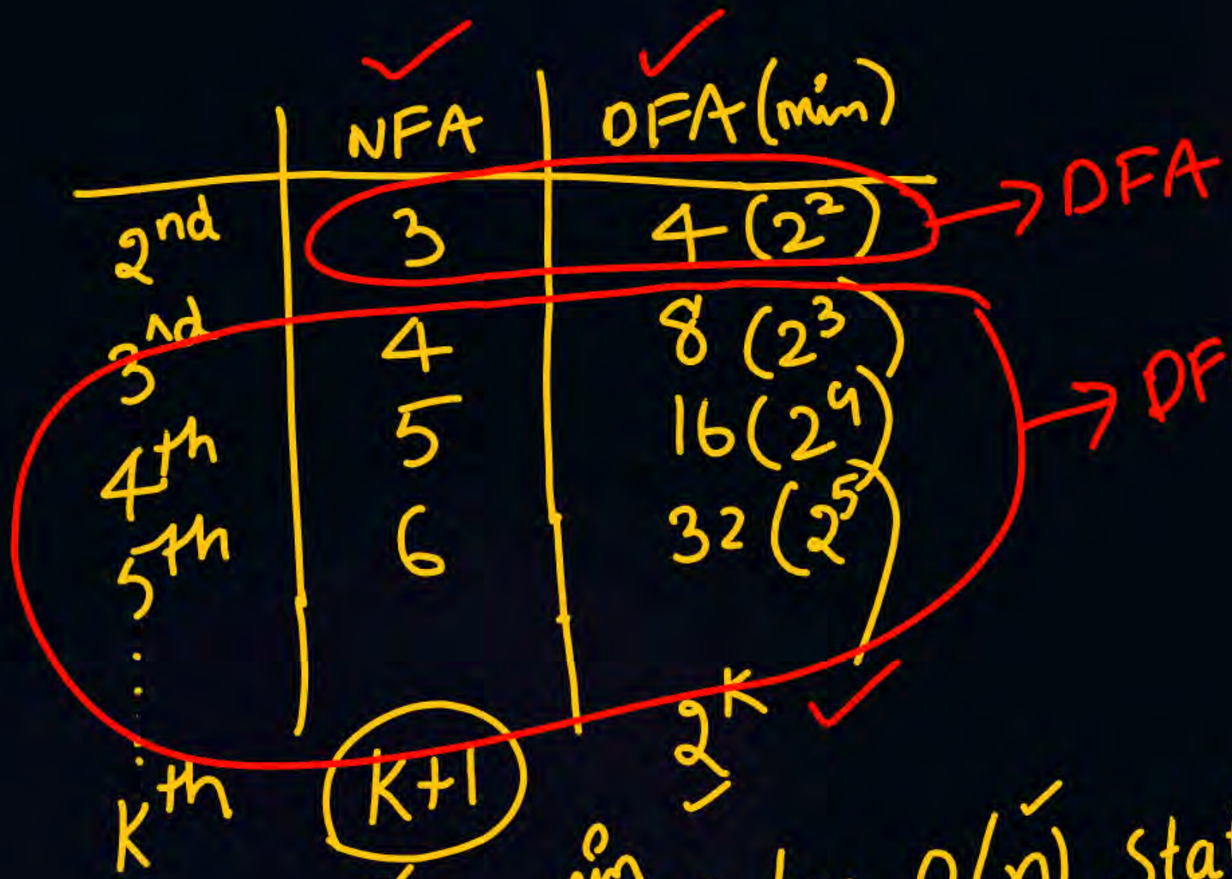
	a	b
→A	A,B	A
B	C	C
C	D	D
*D	—	—

	a	b
→[A]	[AB]	[A]
[AB]	[ABC]	[AC]
[AC]	[ABD]	[AD]
*[AD]	[AB]	[A]
[ABC]	[ABCD]	[ACD]
*[ABD]	[ABC]	[AC]
*[ACD]	[ABD]	[AD]
*[ABCD]	[ABCD]	[ACD]

min DFA

3rd → NFA → 4

DFA → 8 = 2^3



When any NFA has $O(n)$ states, min DFA can have as many as $O(2^n)$ states

THANK - YOU